AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

Faculty of Science and Technology

Project Report

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Course Teacher:	DIPTA JUSTIN GOMES

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	Marks Obtained	
	Total Marks	

Road Safety Public Service Announcement (PSA) Using Computer Graphics

Team Member List

Name	ID	Contribution
QUAZI RESHOAN YAZDI (Corresponding Member)	21-45485-3	Report: Introduction **Feature: Rickshaw Lane Change**
TANVIR MAHTAB TANEEM	23-51975-2	Report: Background to the problem **Feature: Wrong side car driving**
S M KHALED HASAN PRINCE	23-54257-3	Report: Methods Used **Feature: Traffic Light Compliance Simulation**
TUTUL KUMAR GHOSH	22-47813-2	Report: Future direction, Conclusion **Feature: Bus Stop**

Introduction

Dhaka, the vibrant and densely populated capital of Bangladesh, is a city perpetually in motion, yet paradoxically, often at a standstill. Its streets, the lifeblood of the metropolis, pulse with a chaotic energy that is both a testament to its economic vitality and a glaring symptom of a deep-seated crisis. This crisis is twofold: severe, daily traffic congestion that strangles productivity and wastes countless hours, and a pervasive lack of safety that leads to frequent, often devastating, accidents. The problem is not merely one of inadequate infrastructure, but fundamentally one of

human behavior. A casual observer on any major thoroughfare—from the bustling Gulshan Avenue to the historic streets of Old Dhaka—would witness a relentless cycle of traffic rule violations that have become normalized to the point of routine.

Public transport drivers, locked in fierce competition for passengers, routinely bring their massive buses to a halt in the middle of intersections, creating immovable obstacles that trigger cascading gridlock in all directions. Autorickshaws, the ubiquitous three-wheelers that weave through traffic, frequently change lanes or make sudden turns without the simple, life-saving act of checking their side mirrors, oblivious to the vehicles in their blind spots. Motorcyclists, often in a hurry, navigate the chaotic streets with a dangerous sense of invincibility, their helmets—if they own one—more likely to be hanging from the handlebars than protecting their heads. The buses and minibuses themselves are often packed far beyond their intended capacity, with passengers spilling out of doors, compromising the vehicle's stability and the safety of those inside. Compounding these issues is the alarming prevalence of vehicles driven without proper licenses or valid fitness certificates, putting dangerously maintained machines in the hands of unqualified individuals.

Chaos is not confined to those behind the wheel. Pedestrians, faced with a landscape often hostile to those on foot, are forced to become part of the problem. Zebra crossings are frequently ignored, rendered meaningless by a continuous flow of traffic that rarely yields. Instead, people are seen darting across multi-lane roads, calculating risks with every step, their safety left to chance. Over speeding in densely populated areas further exacerbates the danger, turning minor lapses in judgment into fatal collisions. This collective disregard for basic traffic rules creates an environment where the simple act of moving from one point to another becomes a high-stakes gamble.

However, addressing this behavioral crisis is fraught with a unique challenge: literacy. A significant segment of the population involved in the transport sector, including many drivers of buses, autorickshaws, and trucks, as well as a substantial number of pedestrians, are semi-literate or illiterate. This reality renders a vast majority of conventional public awareness campaigns—which rely heavily on written signs, printed pamphlets, and text-heavy advertisements—profoundly ineffective. A written slogan or a list of rules holds little meaning for someone who cannot read it. The crucial message of safety, of cause and effect, fails to cross the literacy barrier. In this context, communication must transcend the written word. It must be immediate, intuitive, and universal. A visual medium, one that can convey complex scenarios and their consequences through imagery alone, is not just an alternative; it is an imperative.

It is at this intersection of a critical social problem and a communication gap that this project finds its purpose. This proposal outlines the creation of a Computer Graphics–based Public Service Announcement (PSA) designed specifically to visually demonstrate safe versus unsafe driving and pedestrian practices in the context of Dhaka. The core methodology involves using the fundamental graphical primitives available in OpenGL and GLUT—such as circles, rectangles, polygons, and lines—to construct a simplified, animated model of the city's traffic ecosystem. This approach will allow for the creation of clear, unambiguous narratives. We will animate stark contrasts: a bus blocking an intersection versus one pulling over correctly; an autorickshaw turning without a mirror check resulting in a collision versus one checking and turning safely; a motorcyclist without a helmet suffering a head injury versus one with a helmet surviving a fall; pedestrians jaywalking causing accidents versus using a zebra crossing safely.

The goal of this project is twofold. Firstly, it aims to create awareness by providing an easy-to-understand visual guide. It will show not just "what not to do," but vividly illustrate the negative outcomes—the collisions, injuries, the massive delays—that stem from these actions. Simultaneously, it will present positive, safe alternatives, demonstrating how simple behavioral changes can lead to safer outcomes for all. Secondly, the project seeks to demonstrate the practical relevance of computer graphics as a discipline. It moves beyond theoretical exercises and abstract rendering to show how this powerful technological tool can be applied to solve a real-life, socio-technical problem in a developing world context like Bangladesh. By using visual storytelling for education, we highlight how technology can be harnessed for social good, creating tools that are accessible to the very people who need them the most.

By the conclusion of this project, we aim to deliver a working graphics simulation that serves as both a functional academic demonstration and a prototype with genuine social impact potential. It will stand as a testament to the idea that computer graphics can be a potent teaching tool, capable of reaching illiterate and semi-literate audiences through the universal language of visuals. In using the logic of pixels and polygons to model the chaos of Dhaka's streets, we hope to contribute, in a small but meaningful way, to a future where those streets are safer and more orderly for every citizen.

Background of the problem:

To understand the daily reality of Dhaka's streets is to understand a complex and frustrating puzzle. The city's infamous traffic congestion is more than just an inconvenience; it is a multifaceted crisis with deep-rooted causes that stretch beyond the simple explanation of "too many cars." While the sheer volume of vehicles is a major factor, the true anatomy of the problem lies in a combination of historical planning failures, powerful economic incentives, a critical communication gap, and a breakdown in the social contract of road use. This background will explore these key areas to establish why a novel, visual approach to public awareness is not just beneficial, but essential.

The Foundation: A City Built for a Different Era

Dhaka’s transformation in the late 20th and early 21st centuries has been nothing short of explosive. What was once a relatively modest city has rapidly expanded into one of the most densely populated megacities on Earth, with a population exceeding 20 million people. This rapid, largely unplanned growth has placed immense strain on an infrastructure that was designed for a different time and a much smaller population.

A critical statistic highlights this fundamental flaw: only about 7-8% of Dhaka's land area is dedicated to roads. For context, many other major global cities allocate 20-30% of their space to transportation networks. This basic scarcity of physical space is the primary constraint within which all other traffic problems occur. There is simply not enough asphalt to efficiently move the millions of people and goods that need to circulate daily. This lack of space is compounded by poor planning. The road network lacks a sufficient number of strategic connectors, flyovers, and underpasses that would allow for the smooth flow of traffic around bottlenecks. Key intersections, often simple at-grade crossings, become critical failure points. When one vehicle stops incorrectly, it doesn't just affect the lane it's in; it creates a ripple effect that can paralyze an entire neighborhood for hours.

Furthermore, the existing road space is constantly under encroachment. Illegal parking, a vibrant but chaotic street vendor economy, and construction materials for the city's perpetual building boom routinely spill onto the streets, effectively reducing two-lane roads to one, and one-lane roads to impassable alleys. This environment means there is no redundancy in the system; when one artery is blocked, there are often no viable alternative routes.

The Human Element: Economic Survival vs. Traffic Rules

If infrastructure is the skeleton of the traffic system, then the behavior of its users is the central nervous system. Here, the situation is driven not by malice, but by a harsh economic reality. A significant portion of Dhaka's public transport system—the buses, minibuses, and autorickshaws that carry the majority of its citizens—operates on a fiercely competitive, privately-owned model. For the drivers and their helpers (known as "khalasis"), income is not a fixed salary but a commission based on the number of fares collected.

This "race for passengers" creates a set of perverse incentives that directly lead to the reckless driving habits that define Dhaka's streets. A driver's livelihood depends on completing as many trips as possible and packing each vehicle to its absolute maximum capacity. This economic pressure manifests in clear, dangerous behaviors:

- **Indiscriminate Stopping:** Buses and minibuses will stop anywhere and everywhere to pick up a waiting passenger, even if it means halting in the middle of a busy intersection or a narrow bridge. The financial gain from one more fare outweighs the collective cost of the traffic jam they create behind them.
- **Over speeding and Racing:** To get to the next bus stop before a competitor, drivers will speed and engage in dangerous maneuvers. The practice of "bus bullying," where rival buses race and cut each other off, is a direct and deadly consequence of this competition.
- **Overloading:** Vehicles are routinely filled far beyond their safe capacity, with passengers hanging out of doors. This compromises vehicle stability and safety but maximizes revenue for that trip.

In this context, obeying traffic rules—waiting for a green light at an empty intersection, properly pulling into a bus bay, or driving at a safe speed—is perceived as a direct financial loss. The system punishes safe drivers by making them earn less. This creates a classic "tragedy of the commons," where individuals acting in their own rational self-interest (maximizing income) collectively create an irrational and dysfunctional system (city-wide gridlock).

The Critical Barrier: When Words Fail

A pivotal, yet often underestimated, dimension of this crisis is the challenge of communication and literacy. While Bangladesh has made significant progress in raising its national literacy rate, the demographic that comprises a large portion of professional drivers—particularly bus, truck, and autorickshaw operators—often includes individuals from lower-income, rural backgrounds who have had limited access to formal education. Many are functionally illiterate or semi-literate, meaning they can perhaps sign their name or recognize numbers, but cannot comprehend paragraphs of text or complex written instructions.

This literacy gap acts as a formidable barrier to conventional public safety campaigns. A written traffic sign that says "No Stopping," "Use Helmet," or "Yield" is nothing more than a pattern of

shapes to someone who cannot read it. Government pamphlets, detailed newspaper articles, and text-heavy television scrolls about road safety fail to penetrate this crucial demographic. The law exists in thick rulebooks, but its principles do not translate into understanding or behavioral change on the street. Consequently, rules are often broken not out of defiance, but out of a lack of comprehension of the rule itself and, more importantly, a failure to grasp the potentially fatal *consequence* of breaking it. An autorickshaw driver may not have been taught the physics of a blind spot; he has never been *shown*, in a clear and impactful way, how a sudden turn without a mirror check can lead to a catastrophic collision.

The Collapse of Enforcement and Civic Responsibility

The pervasive culture of rule-breaking is further sustained by a systemic failure of consistent and perceived-fair enforcement. The traffic police force, while dedicated, is critically understaffed and underequipped to manage the scale of the problem. Their presence is often limited to major intersections, leaving vast stretches of road lawless. This sporadic enforcement fosters a culture of impunity, where drivers learn the specific locations where they might be penalized and adhere to rules only at those points, reverting to reckless behavior immediately after.

While the government is actively working to combat it, the lingering perception of corruption also undermines the rule of law. The belief that a traffic fine can be settled with a small, informal payment rather than a formal ticket and mandatory safety course erodes the deterrent effect. When the consequence of a violation is negotiable, the behavior is not corrected, and the law loses its moral authority. This leads to a broader erosion of civic responsibility, where the shared space of the road becomes a free-for-all, and the sense that everyone is responsible for collective safety is lost.

The Pedestrian's Plight: A System That Forgets the Person on Foot

Finally, the problem is severely compounded by the complete neglect of the pedestrian. Dhaka is an intensely hostile environment for anyone on foot. Footpaths, where they exist, are often unusable, occupied by shops, street vendors, or parked motorcycles. Zebra crossings are frequently faded, placed in illogical locations, or simply ignored by motorists who have no legal obligation to stop. The limited number of footbridges are often spaced too far apart to be practical for daily life.

Faced with these insurmountable obstacles, pedestrians are forced to become part of the problem. They jaywalk across multi-lane highways, weaving through moving traffic because the designed environment provides no safe, convenient, or dignified alternative. This not only puts their lives at extreme risk every day but also actively contributes to the congestion, as vehicles must constantly slow down and swerve to avoid them.

In conclusion, the traffic chaos in Dhaka is a perfect storm of systemic failures. It is a problem born from inadequate physical space, sustained by economic structures that reward dangerous behavior, exacerbated by a critical communication gap that leaves a key part of the population out of the safety conversation, and perpetuated by weak enforcement and a neglected pedestrian infrastructure. While long-term mega-projects like the Metro Rail are vital for the future, there is a desperate and immediate need for targeted interventions that can alter dangerous behaviors at a mass scale. It is within this context that a visual, universally understandable Public Service Announcement becomes a powerful and necessary tool. By transcending the barrier of literacy and

using the intuitive language of graphics to demonstrate cause and effect, such an initiative can serve as a crucial stopgap, educating and persuading where words and enforcement alone have failed.

Methods used:

#	Function Name	Description
1	glutInit()	Initializes the GLUT library.
2	glutInitDisplayMode()	Sets the display mode (single/double buffering, RGB).
3	glutInitWindowSize()	Sets the initial window size.
4	glutInitWindowPosition()	Sets the initial window position.
5	glutCreateWindow()	Creates a window with the specified title.
6	glutDisplayFunc()	Registers the display callback function.
7	glutTimerFunc()	Registers a timer callback function.
8	glutMouseFunc()	Registers the mouse callback function.
9	glutKeyboardFunc()	Registers the keyboard callback function.
10	glutPostRedisplay()	Marks the current window for redisplay.
11	glutSwapBuffers()	Swaps the front and back buffers (used with double buffering).
12	glutMainLoop()	Enters the GLUT event processing loop.
13	glutGet()	Returns state information about GLUT.
14	glutBitmapCharacter()	Renders a character using a bitmap font.
15	glClear()	Clears buffers to preset values.
16	glClearColor()	Specifies clear values for color buffers.
17	glLoadIdentity()	Replaces the current matrix with the identity matrix.
18	glMatrixMode()	Specifies which matrix is the current matrix.
19	glOrtho() / gluOrtho2D()	Sets up a 2D orthographic viewing region.
20	glPushMatrix()	Pushes the current matrix onto the stack.
21	glPopMatrix()	Pops the top matrix off the stack.
22	glTranslatef()	Multiplies the current matrix by a translation matrix.
23	glScalef()	Multiplies the current matrix by a scaling matrix.

24	glColor3f() / glColor3ub()	Sets the current color (RGB).
25	glColor4f() / glColor4ub()	Sets the current color (RGBA).
26	glBegin()	Delimits the vertices of a primitive.
27	glEnd()	Ends primitive specification.
28	glVertex2f()	Specifies a vertex (2D float coordinates).
29	glLineWidth()	Specifies the width of rasterized lines.
30	glPointSize()	Specifies the size of rasterized points.
31	glBegin(GL_POINTS)	Starts drawing point primitives.
32	glBegin(GL_LINES)	Starts drawing line primitives.
33	glBegin(GL_LINE_LOOP)	Starts drawing connected line segments.
34	glBegin(GL_POLYGON)	Starts drawing a filled polygon.
35	glBegin(GL_TRIANGLES)	Starts drawing triangle primitives.
36	glBegin(GL_TRIANGLE_FAN)	Starts drawing triangle fan primitives.
37	glRasterPos2f()	Sets the raster position for bitmap text.
38	glEnable()	Enables a server-side OpenGL capability (e.g., GL_BLEND).
39	glDisable()	Disables a server-side OpenGL capability.
40	glBlendFunc()	Specifies pixel arithmetic for blending.

Feature set:

Wrong side car driving

This OpenGL-based traffic safety simulation operates through an interactive graphical system where users select between "TO DO" (safe) and "NOT TO DO" (dangerous) driving scenarios via mouse-clickable buttons. When activated, the "TO DO" scenario animates two vehicles moving synchronously in the same direction using continuous coordinate translation, demonstrating proper lane discipline. Conversely, the "NOT TO DO" scenario programs vehicles to approach from opposite directions, with collision detection triggered when their transformed positions intersect. Upon collision, the system generates a dynamic fire animation using blended translucent circles that progressively scale and fade, visually emphasizing the consequences of reckless driving. The environment features automatically scrolling clouds through position updates, while buildings, trees, and road elements constructed from GL primitives create an immersive urban context. Real-time state management handles scenario resetting, smoothly returning vehicles to starting positions after the fire sequence is completed, and updates instructional messages to reinforce the safety lesson through direct visual comparison of outcomes.

Traffic Light Compliance Simulation

This OpenGL-based traffic safety simulation creates an interactive urban road environment where users can demonstrate key safety principles through keyboard-controlled scenarios, emphasizing helmet usage and traffic light compliance. The system renders a detailed scene with GL primitives for roads, footpaths, fences, shops, houses, and animated elements like scrolling clouds and dynamic vehicles. Four bikes are programmed with position translation updates via timer functions, allowing horizontal and vertical movement along designated lanes. Users toggle traffic signals using 'R' for red (halting horizontal traffic) and 'G' for green (allowing it), with light states visualized through colored circles and glow effects using GL blending for realistic illumination. Helmet functionality is activated via 'H', rendering a protective layer on riders' heads; without it, collision detection between opposing bikes (triggered by 'A' or proximity checks) initiates an accident sequence, animating a falling rider with coordinate offsets and displaying warning messages. A traffic police figure, positioned near the road, can be activated with 'O' to blink a sticklight (using counter-based toggling) and show overspeeding alerts, reinforcing enforcement themes. Resetting occurs with 'X', returning elements to initial states. The simulation educates through visual contrast: safe practices (helmet on, signals obeyed) allow smooth traffic flow, while violations trigger dramatic consequences, with instructional text overlays using raster positioning to highlight rules like "Press O to toggle police" in colored bitmap fonts for enhanced user guidance and safety awareness.

Rickshaw on the wrong lane:

The Rickshaw on the wrong lane Simulation is a computer graphics project developed using OpenGL and GLUT in C++, designed to visualize and demonstrate unsafe road behavior through an interactive 2D animated environment. The primary objective of this project is to illustrate a real-world traffic safety issue where a rickshaw overtakes a car in the fast lane, compelling the car to decelerate and consequently disrupting the normal flow of traffic. The simulation aims to raise awareness regarding lane discipline, safe driving practices, and overall road safety, aligning with the broader objective of using technology to promote social responsibility.

The project environment consists of a multi-lane road scene, designed with appropriate background elements such as dividers, trees, and a skyline to create a realistic visual context. The moving vehicles are individually modeled and animated using frame-based motion and geometric transformations (translation, scaling, and rotation) to produce smooth and continuous animation sequences. The timing and synchronization between the car and the rickshaw are precisely managed to depict the overtaking incident effectively and realistically. Through the use of color modeling and layering techniques, the project achieves visual clarity and scene depth, enhancing the overall viewing experience.

From a technical standpoint, the project applies key concepts of computer graphics and animation, including the use of double buffering for flicker-free rendering, hierarchical modeling for structured object creation, and perspective management for appropriate scene scaling.

Furthermore, the program supports user interaction through keyboard inputs, enabling the user to pause, restart, or modify the animation speed as desired. This interactive functionality adds value by allowing controlled demonstrations and replays, making the simulation suitable for both academic and awareness-based presentations. The program structure follows a modular design

approach, separating drawing routines, event handling, and animation logic to ensure maintainability and scalability of the codebase.

Overall, the Rickshaw on the wrong lane Simulation serves as an educational and technically comprehensive computer graphics application. It effectively integrates theoretical graphics principles with a real-world safety message, demonstrating how programming and visual design can be combined to address social issues. By presenting a realistic scenario of unsafe overtaking behavior, the project contributes both to academic learning in the field of computer graphics and to the promotion of responsible road usage through visual education.

Bus Stop

This Bus Stop simulation offers a rich interactive environment centered around public transportation, featuring a dynamic day/night cycle toggle and multi-scene navigation system that allows players to explore different urban settings. The core experience involves coordinated vehicle control, including bus navigation via arrow keys, boat activation with left-click, and car movement with right-click, all converging around the bus stop hub. The simulation includes character animation systems with a walking "vodka man" controlled by keyboard commands and sophisticated passenger mechanics where a girl character can board at the bus stop and exit at designated destinations like the university. The urban landscape showcases detailed transportation infrastructure with road networks featuring yellow markings, a dual-track railway bridge, and multiple architectural structures that create a believable transit environment. Built on OpenGL graphics rendering with efficient state management, the project provides intuitive mouse and keyboard controls for managing bus operations, smooth animation coordination between multiple moving objects, and educational insights into public transportation systems and urban mobility concepts through its modular, scalable code architecture that creates an engaging bus stop ecosystem and transit experience.

Future Direction:

The Bus Stop simulation platform has significant potential for expansion into a comprehensive traffic safety and public transportation education suite. Future development could integrate the wrong-side driving, traffic light compliance, and wrong-lane rickshaw scenarios directly into the bus route environment, creating unified urban safety narratives. Enhanced collision detection systems could be implemented across all vehicle types, while pedestrian safety modules could be added showing proper bus stop behavior and crosswalk usage. Multiplayer functionality would allow role-playing as different traffic participants (driver, pedestrian, traffic police), and data logging could track decision-making patterns for educational assessment. Mobile deployment with touch interfaces, VR integration for immersive perspective-taking, and real-world scenario customization based on local traffic patterns would further enhance its educational impact. Additional modules could include weather-based driving challenges, emergency vehicle right-of-way situations, and public transportation etiquette scenarios, transforming the platform into a complete urban mobility education tool.

Conclusion:

The Bus Stop simulation successfully demonstrates how interactive computer graphics can effectively communicate crucial transportation safety concepts through engaging visual

storytelling. By combining the wrong-side driving prevention, traffic light compliance, lane discipline awareness, and public transportation education into a cohesive framework, the project creates a powerful educational tool that bridges technical computer graphics principles with real-world social responsibility. The modular architecture allows for continuous expansion while maintaining intuitive user interaction through both keyboard and mouse controls. This project not only serves as an academic demonstration of OpenGL programming excellence but also as a meaningful contribution to traffic safety awareness, showing how technology can visualize consequences of dangerous behaviors while promoting safer transportation habits. The simulation stands as a testament to how computer graphics can transcend entertainment to address critical societal issues, making it equally valuable for computer science education and public safety advocacy.

Implementation code:

Code implementation is provided in the GitHub link: **Traffic-PSA**

Screenshot of the Project:

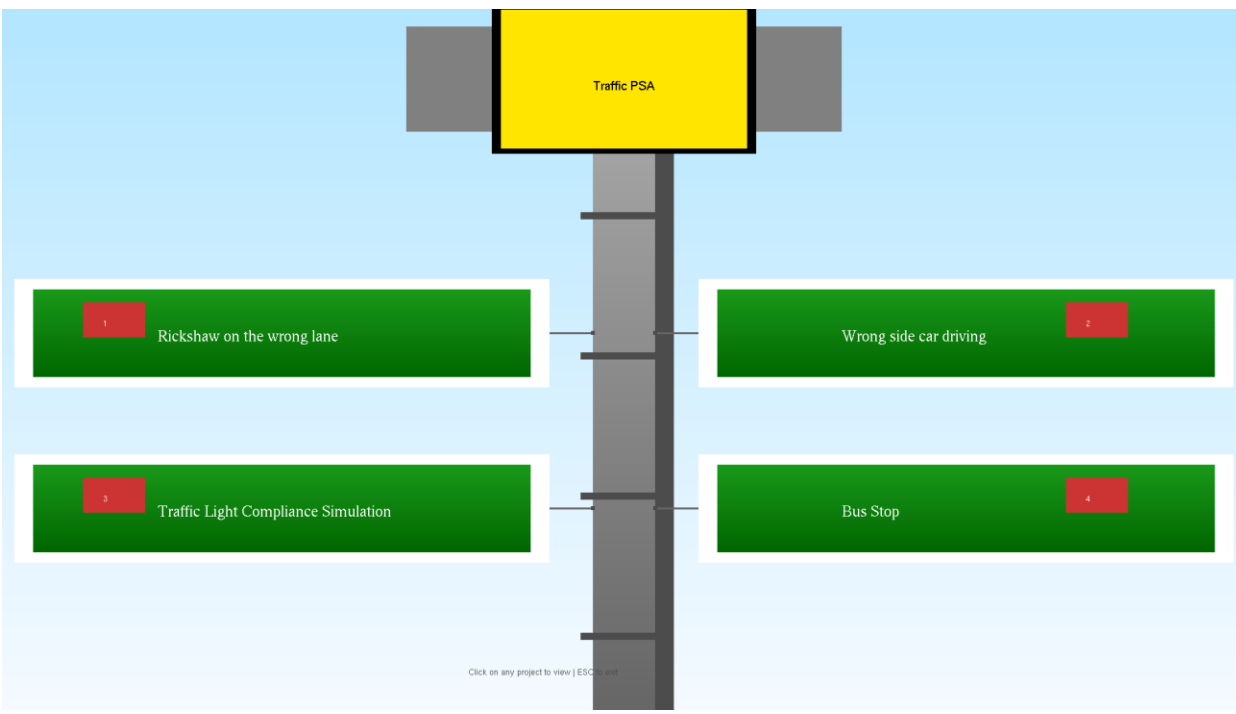

Figure 1: Menu

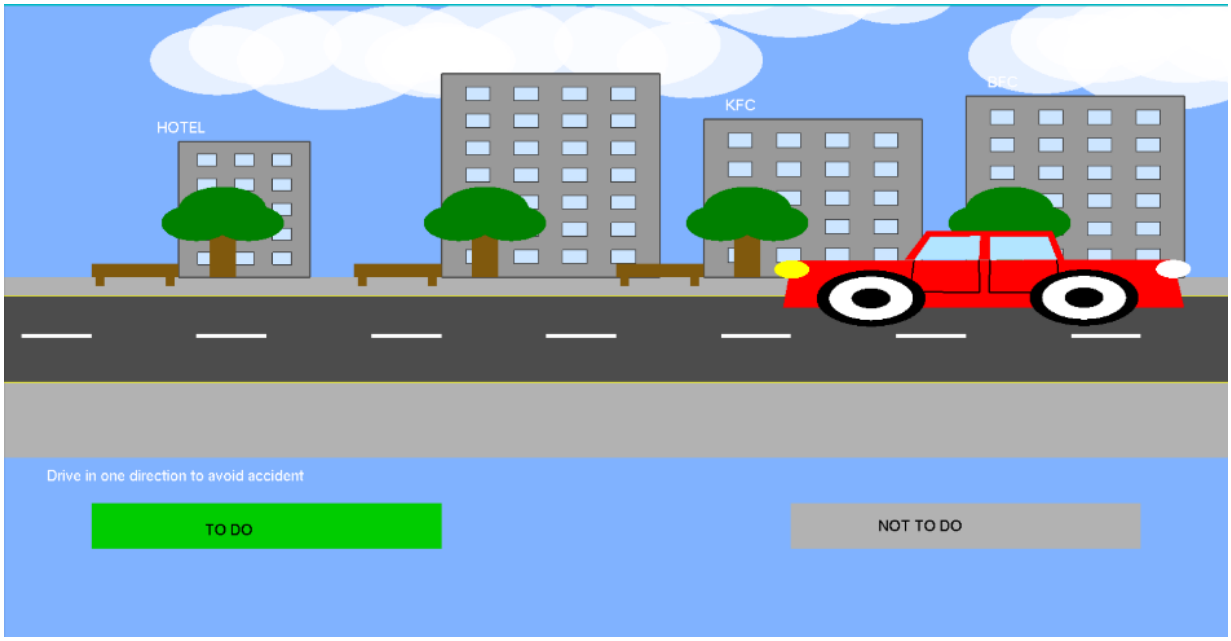

Figure 2. Right side car driving

Figure 3. Wrong side car driving

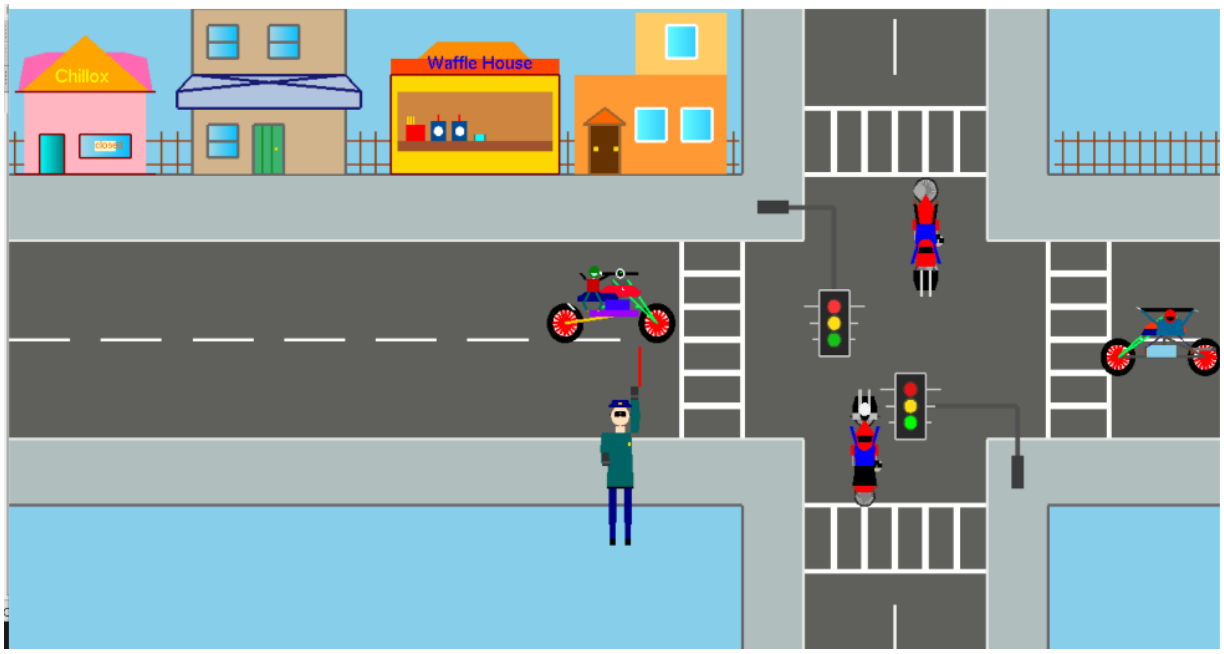

Figure 4. Wrong Corrected Road Scene with Green Signal

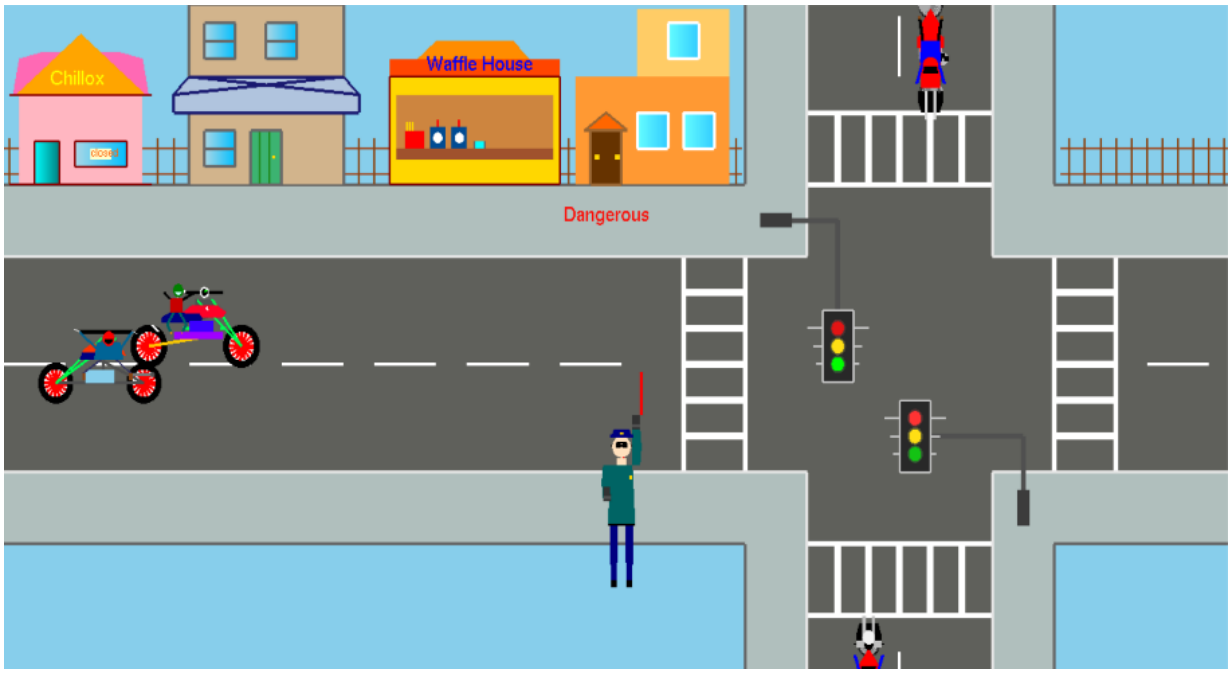

Figure 5. Red Signal Violation and Overspeeding

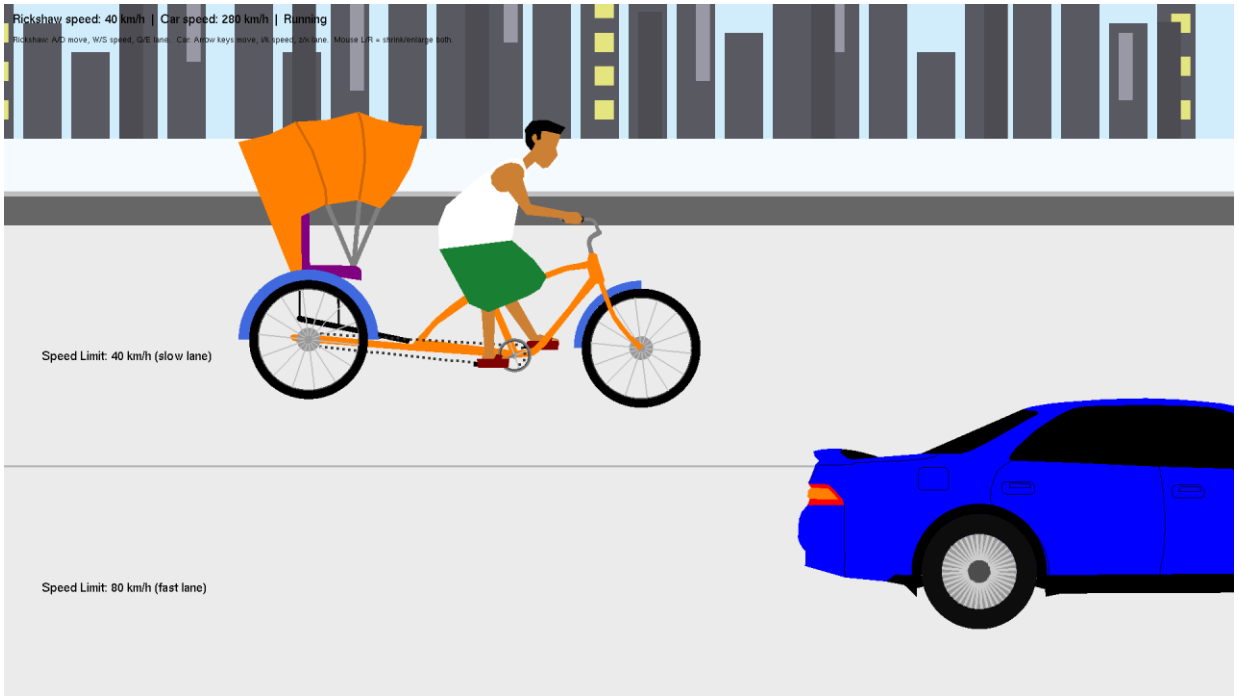

Figure 6. Rickshaw in the right lane

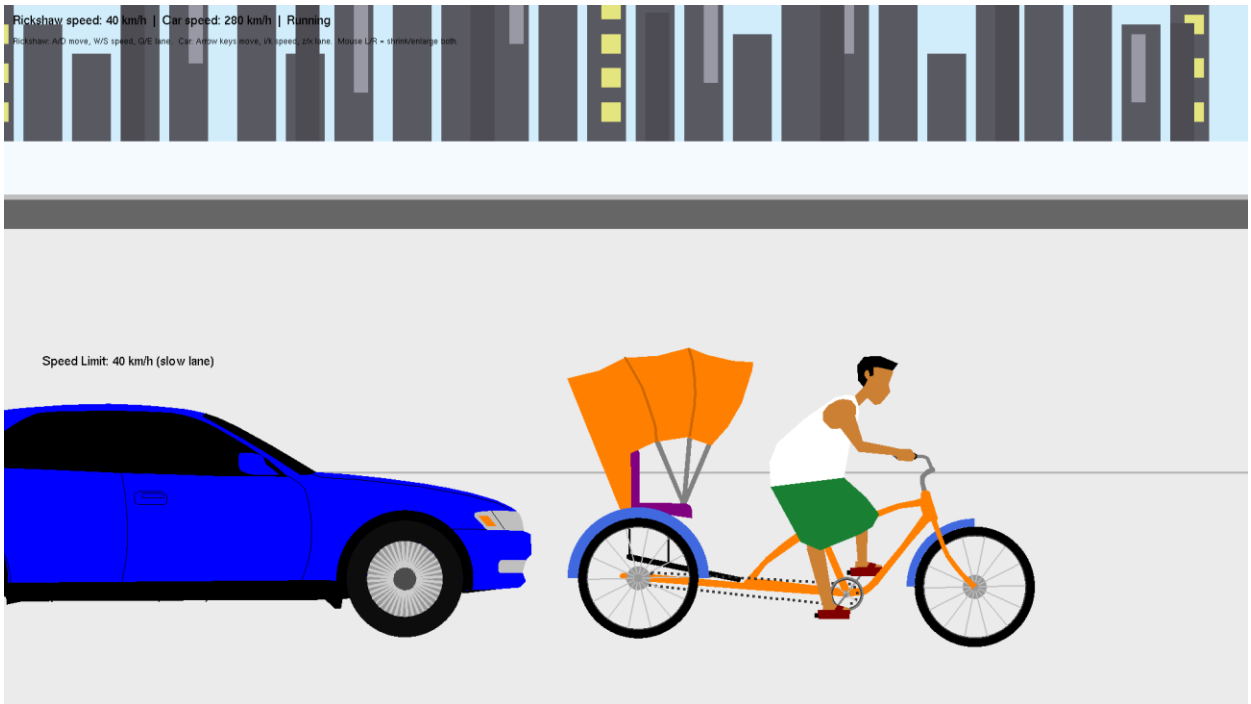

Figure 7. Rickshaw in the wrong lane.

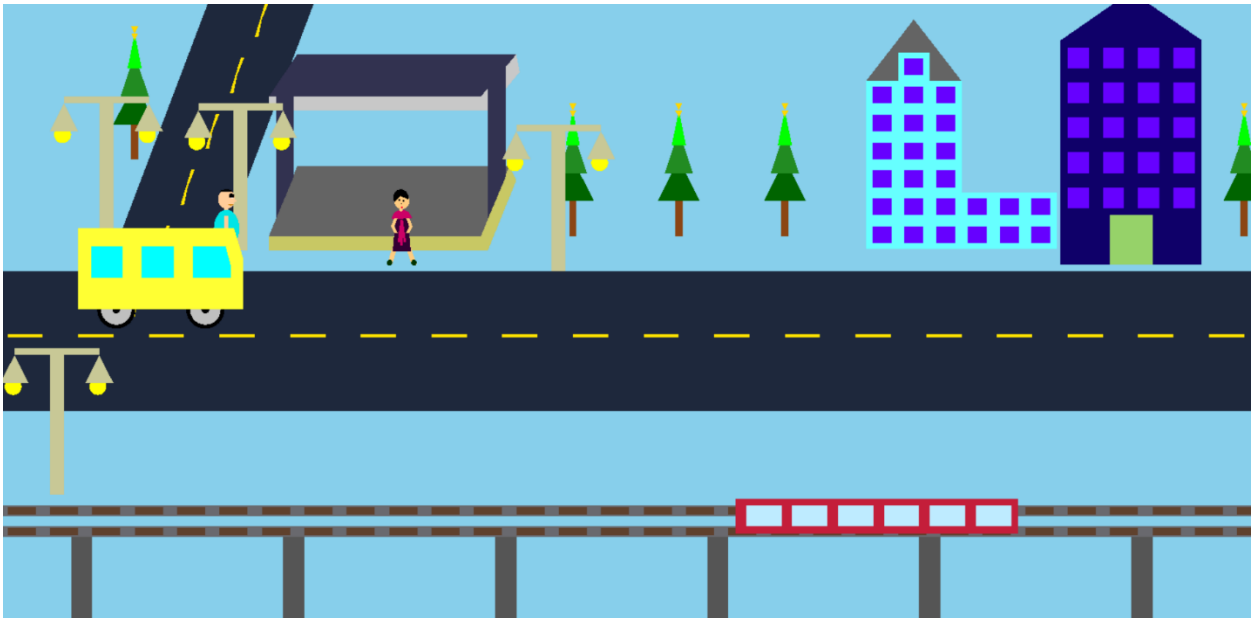

Figure 8. Pepole waiting for bus

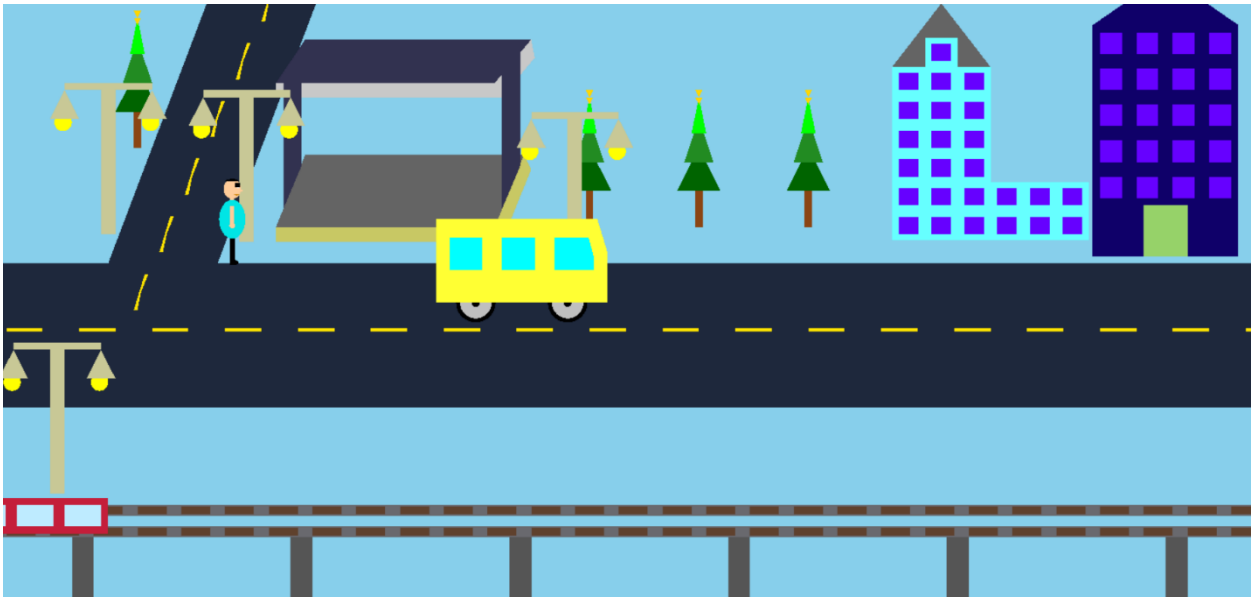

Figure 9.Getting into the bus

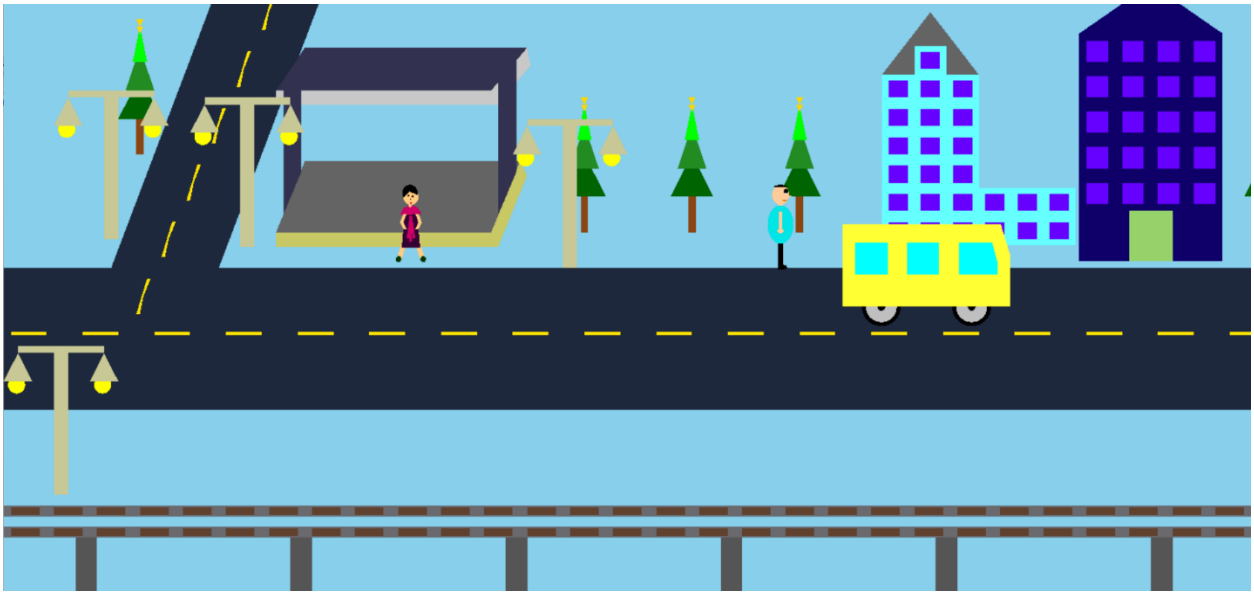

Figure 10. Man walking on the road

GitHub link:

Traffic-PSA

List of references:

- **OpenGL-Traffic-Simulation**: https://github.com/AsifurRahmanTanvir/OpenGL-Traffic-Simulation
- **Dhaka-Traffic-Simulation**: https://github.com/Nazmul28/Dhaka-Traffic-Simulation
- **Traffic-Management-System**: https://github.com/rimon50/Traffic-Management-System